

Challenge (Habanero)

- Q1.** Solve the system using substitution: $x + 2y = 7$, $3x - 2y = 5$. What is x ?
- (A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q2.** A gamer's high score is determined by $s = 2x + 5y$, where x is the number of levels completed and y is the number of coins collected. If $s = 27$ and $x = 4$, what is y ?
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q3.** Solve the system using elimination: $4x - 3y = 2$, $2x + 3y = 8$. What is y ?
- (A) -1
(B) 0
(C) 1
(D) 2
(E) 3
- Q4.** A game developer has 240 minutes to test a new game. The testing time is determined by $t = 4x + 2y$, where x is the number of levels and y is the number of characters. If $x = 30$ and $t = 160$, what is y ?
- (A) 10
(B) 15
(C) 20
(D) 25
(E) 30
- Q5.** What is the solution to the system $x - 2y = -3$, $-x + 2y = 1$?
- (A) No solution
(B) Infinitely many solutions
(C) (1, 2)
(D) (2, 1)
(E) (0, 0)
- Q6.** A coding team has 12 hours to complete a project. The project time is determined by $t = 2x + 3y$, where x is the number of modules and y is the number of tests. If $t = 9$ and $x = 2$, what is y ?
- (A) 1
(B) 1.5
(C) 2
(D) 2.5
(E) 3
- Q7.** Solve the system using substitution: $2x - 5y = -1$, $x + 2y = 9$. What is y ?
- (A) -1
(B) 0
(C) 1
(D) 2
(E) 3
- Q8.** What is the solution to the system $3x + 2y = 7$, $-3x - 2y = -7$?
- (A) No solution
(B) Infinitely many solutions
(C) (1, 2)
(D) (2, 1)
(E) (0, 0)

Open Ended Questions (FRQ)

- Q9.** Solve the system of equations using elimination: $x + 4y = 10$, $3x - 2y = 5$. Show all steps and box the final answer.
- Q10.** A gaming company tracks player data using the system: $p = 2x + 3y$, $q = x - 2y$, where p is the player's score and q is the player's level. Solve for x and y when $p = 20$ and $q = 5$. Show all steps and box the final answers.

Chef's Tips

- Ensure students show all steps for free-response questions
- Encourage students to use substitution and elimination methods interchangeably
- Remind students to check their solutions by plugging them back into the original equations